

Applications of Graph Theory

Facts.

- The sum of degrees of all vertices is equal to two times the number of edges.
- A connected graph of n vertices has at least $n - 1$ edges.
- A connected graph of n vertices is a tree (i.e. having no cycle) if and only if it has $n - 1$ edges.
- A graph has no odd cycle if and only if it is a bipartite graph.
- (Mantel's theorem) A graph of n vertices and more than $\frac{n^2}{4}$ edges must have a triangle.
- $R(3, 3) = 6$.
- An undirected connected graph has an Eulerian circuit if and only if all vertices have even degree.
- An undirected connected graph has an Eulerian trail if and only if only 0 or 2 vertices have odd degree.
- A directed and strongly connected graph has an Eulerian circuit if and only if every vertex has equal in-degree and out-degree.
- (Hall's theorem) A bipartite graph with bipartite sets X and Y has a matching that covers every vertex in X if and only if $|W| \leq |f(W)|$ for every subset W of X , where $f(W)$ is the set of all vertices which are adjacent to some vertex in W .

Problems.

Problem 1. Let m and n be positive integers. Some squares of an $m \times n$ board are coloured red. A sequence $a_1, a_2, \dots, a_{2r}$ of $2r \geq 4$ pairwise distinct red squares is called a *bishop circuit* if for every $k \in \{1, \dots, 2r\}$, the squares a_k and a_{k+1} lie on a diagonal, but the squares a_k and a_{k+2} do not lie on a diagonal (here $a_{2r+1} = a_1$ and $a_{2r+2} = a_2$).

In terms of m and n , determine the maximum possible number of red squares on an $m \times n$ board without a bishop circuit.

(Remark. Two squares lie on a diagonal if the line passing through their centres intersects the sides of the board at an angle of 45° .)

Problem 2. Let a and b be positive integers, and let A and B be finite sets of integers satisfying

- (i) A and B are disjoint;
- (ii) if an integer i belongs to either to A or to B , then either $i + a$ belongs to A or $i - b$ belongs to B .

Prove that $a|A| = b|B|$. (Here $|X|$ denotes the number of elements in the set X .)

Problem 3. Consider any triangulation T of a convex polygon $\mathcal{P}$ with 100 sides. For each diagonal AC of $\mathcal{P}$, we say that it is *good* if there exist two more vertices B and D of $\mathcal{P}$ such that $ABCD$ is convex, and all the segments AB, BC, CD, DA, BD are drawn in T . Find the maximum number of pairs of good diagonals which intersect inside $\mathcal{P}$.

Problem 4. Let n be a positive integer. Prove that there exists a binary string $a_1a_2 \cdots a_{2^n}$ of length 2^n (where $a_j = 0, 1$ for each j) such that for every binary string $b_1b_2 \cdots b_n$ of length n (where $b_j = 0, 1$ for each j), there exists k such that $a_{k-1+j} = b_j$ for $j = 1, 2, \dots, n$ (where the indices of a_i are taken modulo 2^n).

Problem 5. Given that $a_1, a_2, \dots, a_{2020}$ are integers, find the maximal number of subsequences $a_i, a_{i+1}, \dots, a_j$ ($0 < i \leq j < 2021$) with sum 2021.

Problem 6. Let T be a finite set of positive integers greater than 1. A subset S of T is called *good* if for every $t \in T$ there exists some $s \in S$ with $\gcd(s, t) > 1$. Prove that the number of good subsets of T is odd.